

William Stanley *(1858-1916)* was an inventor and engineer. He developed the first practical **transformer** (which spurred the development of AC power) as well as other developments; like an improved **electric meter** and the first metal **thermos bottle** (vacuum flask). He lived most of his life and ran his businesses in Western Massachusetts during the golden age of electric development.

Stanley is one of the most important inventors of AC power, his work puts him on a par with Edison, Westinghouse, Tesla, Elihu Thomson, Charles Brush and other major personalities of the time. Like the other great pioneers of the time he loved to tinker and design, held a disdain for legal quagmires of patent proceedings, and had his own set of friends and rivals in the industry.

(I_o) is quite small compared to rated current of the transformer, the voltage drop due to this current that can be taken as negligible. As the transformer is open circuited, there is no output. Hence the input power consist of core losses and copper loss in transformer during no load condition.

6.7.2 Short Circuit Test on Transformer

The short circuit test on transformer is used to determine copper loss in transformer.

The connection diagram for short circuit test on transformer is shown in Figure 6.12. A voltmeter, wattmeter, and an ammeter are connected in primary side of the transformer as shown. A reduced voltage at rated frequency is applied to the primary side with the help of a variac of variable ratio auto transformer.

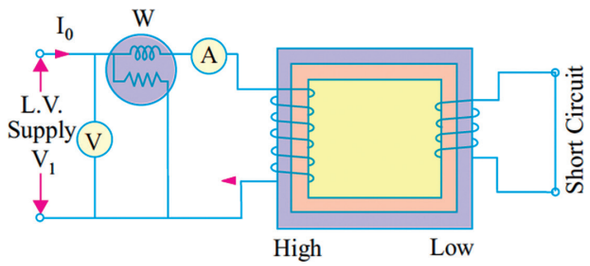


Fig 6.12 *Short Circuit Test*

The secondary side of the transformer is short circuited. Now with the help of variac applied voltage is slowly increased until the ammeter gives reading equal to the rated current in the primary side. Then three instruments reading (voltmeter, ammeter and watt-meter) are recorded. The voltmeter reading is very small compared to the rated primary voltage of the transformer. Here the core losses in transformer can be taken as negligible.

Let the, voltmeter reading is $V_{sc.}$. The input power during test is indicated by wattmeter reading. As the transformer is short circuited, there is no output, hence the input power consist of copper loss in transformer.

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